

MCard-StarterKit V1 BOM Selected Direction

STARTER / REVIEW MATERIAL ONLY. This is not an approved production BOM or purchase authorization. Exact MPN, footprint, pinout, availability, lifecycle, and JLCPCB assembly status require human verification.

Source: hardware/v1/BOM_SELECTED_V1.md **Export date:** 2026-06-12 **Branch:** feature/v1-kicad-schematic-scaffold

Stock and pricing are observations from tsci search --jlcpcb on 2026-06-12, not a purchase guarantee. Recheck every line immediately before ordering.

Block	Selected direction	Package / size	LCSC observation	Fit and rationale	Blocking verification
MCU/BLE	ESP32-S3-MINI-1-N4R2	SMD module; planning body 15.4 x 20.5 mm	C3013941, extended, 5,196 shown	Small module, 2 MB PSRAM for frame buffers; rotate antenna toward upper-right edge	Official land pattern/height, GPIO/strap allocation, Japan certification applicability
Asset flash	W25Q128JVSQ	SOIC-8 208 mil	C97521, basic, 110,435 shown	16 MB local asset option; basic part and hand-reworkable	Confirm secondary-flash ESP-IDF architecture; omit if module flash is sufficient
Display	EastRising ER-TFT020-3	34.6 x 47.8 x 2.0 mm panel	Off-board/manual assembly	Confirmed supplier dimensions, 240 x 320, ST7789, 4-wire SPI	Order drawing/sample, tail geometry, backlight current, lifecycle
Display connector	FH12-14S-0.5SH(55)	14 pin, 0.5 mm	C5378715, extended, 689 shown	Matches supplier-listed 14-pin connector option	Contact side, mated height, exact footprint and FPC thickness
Dynamic NFC	ST25DV04KC-IE6S3	SO-8	C3304276, extended, 18,587 shown	Larger package simplifies prototype inspection and rework	Order-code/package pinout, RF tuning network, energy-harvest use decision
NFC fallback	NT3H2111W0FHKH	XQFN-8 1.6 x 1.6 mm	C710403, extended, 7,510 shown	Alternate dynamic tag family	Tiny package assembly yield and antenna retuning
USB-C	GCT USB4105-GF-A	Horizontal SMT	C3020560, extended, 295 shown	KiCad library footprint exists; offset bottom-right fit	Exact suffix, shell height, drawing and JLC assembly support
Charger/power path	BQ24074RGTR	QFN-16 EP 3 x 3 mm	C54313, extended, 736 shown	Power-path operation supports use while charging	Thermal calculation, ILIM/ISET, battery NTC strategy, exposed-pad layout
3.3 V rail	TPS63031DSKR	SON-10 EP 2.5 x 2.5 mm	C15516, extended, 2,740 shown	Buck-boost keeps 3.3 V across LiPo discharge	Peak ESP32/display load, inductor choice, efficiency and stability
RGB side lights	SK6805-EC15, multiple per side as needed	1.5 x 1.5 mm side emitter	C2890035, extended, 161,838 shown	Bilateral side emission into transparent-resin light guides	Logic threshold at 3.3 V, quantity/current limit, optical isolation and thermal/brightness test
Buttons	SKRSPACE010	4.2 x 3.2 mm SMD	C139797, extended, 156,252 shown	Stock is stronger than the thinner Panasonic candidate	Verify actuator direction/height; find lower-profile alternate if stack fails
Piezo	PKLCS1212E4001-R1	about 12.2 x 12.2 x 3.0 mm	C113159, extended, 4,632 shown	Passive SMD sounder and known mechanical envelope	Drive waveform/voltage, acoustic port and actual package metadata
Vibration driver	AO3400A	SOT-23	C20917, basic, 1,345,927 shown	Common low-side switch; DNP in baseline	Exact manufacturer/order code, gate resistor, flyback strategy and motor current
USB ESD	TPD2EUSB30DRTR	SOT-723	C3040102, extended, 4,708 shown	Low-capacitance USB data protection	LCSC package metadata is inconsistent across listings; verify footprint and pinout
Battery	Protected LiPo 402025, 150-165 mAh class	4.0 x 20 x 25 mm, rotated	Not an LCSC PCBA part	Fits lower rear zone without overlapping the display	Exact supplier, PCM/wires, swelling, shipping/safety documents, runtime

Not selected for the baseline

- ESP32-S3-WROOM-1-N16R8 remains a memory-rich fallback, but consumes more display/RF edge area.
- ESP32-S3-MINI-1U is not baseline because an external antenna adds enclosure and certification variables.
- Vibration motor remains DNP until stack-height and inrush tests pass.
- Aluminum rear parts remain deferred to V2/V3.

Clean-room public-safe design material. No vendor firmware, cloud endpoint, captured application code, private identifier, or proprietary asset is included.